

Lecture 10 - Multiple Regression

Bill Perry

Lecture 09: Review

Covered

- Regression T-Test Anova
- Regression Assumptions
- Model II Regression

Lecture 10: Overview

Multiple Linear Regression model

- Regression parameters
- Analysis of variance
- Null hypotheses
- Explained variance
- Assumptions and diagnostics
- Collinearity
- Interactions
- Dummy variables
- Model selection
- Importance of predictors

Lecture 10: Analyses

What if more than one predictor (X) variable?

- If predictors continuous
- Mix between categorical and continuous
- Can use multiple linear regression

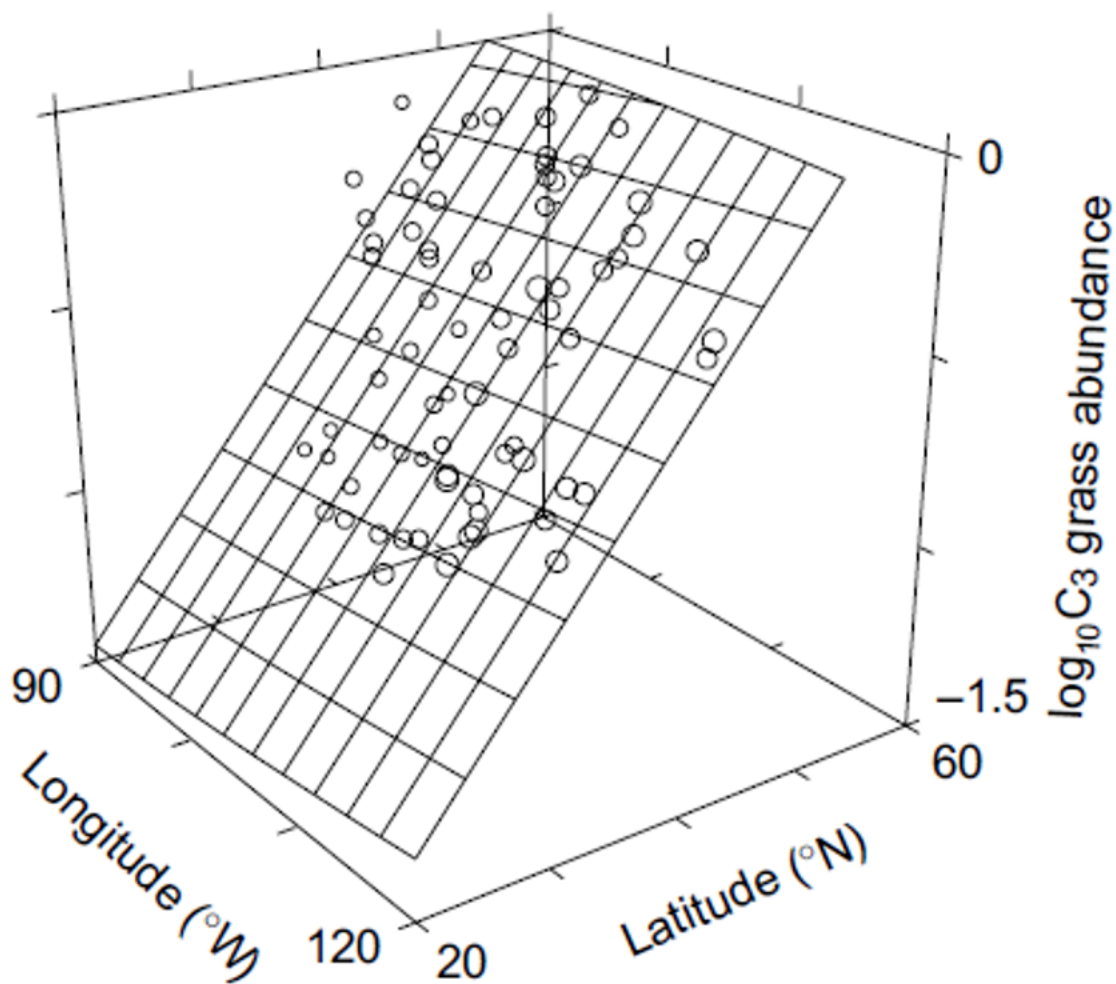
Dependent variable	Independent variable	
	Continuous	Categorical
Continuous	Regression	ANOVA
Categorical	Logistic regression	Tabular

Lecture 10: Analyses

Abundance of C3 grasses can be modeled as function of

- latitude
- longitude
- both

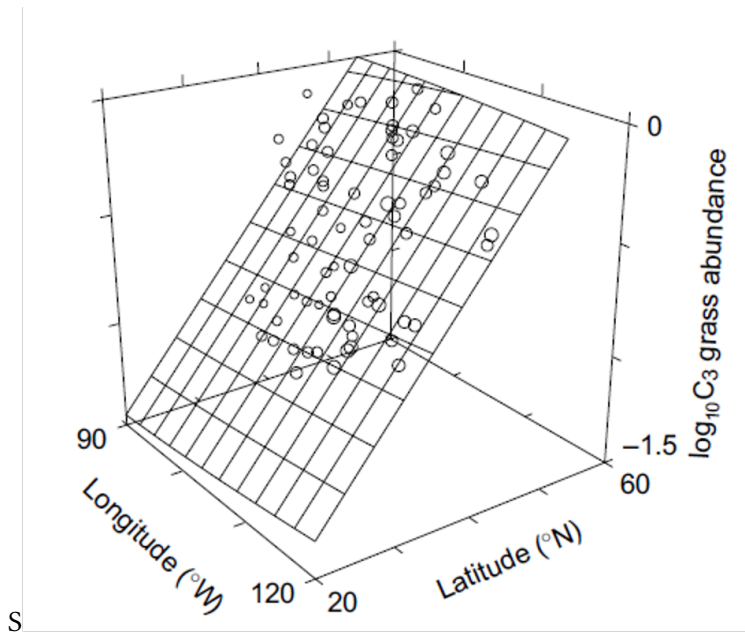
Instead of line, modeled with (hyper)plane



Lecture 10: Analyses

Used in similar way to simple linear regression:

- Describe nature of relationship between Y and X's
- Determine explained / unexplained variation in Y
- Predict new Ys from X
- Find the "best" model



Lecture 10: Analyses

Crawley 2012: “Multiple regression models provide some of the most profound challenges faced by the analyst”:

- Overfitting
- Parameter proliferation
- Multicollinearity
- Model selection



Lecture 10: Analyses

Multiple Regression:

- Set of $i = 1$ to n observations
- fixed X-values for p predictor variables ($X_1, X_2, \dots, X_p$)
- random Y-values:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i$$

- y_i : value of Y for i th observation $X_1 = x_{i1}, X_2 = x_{i2}, \dots, X_p = x_{ip}$
- β_0 : population intercept, the mean value of Y when $X_1 = 0, X_2 = 0, \dots, X_p = 0$

Lecture 10: Multiple linear regression model

Multiple Regression:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i$$

- β_1 : partial population slope, change in Y per unit change in X_1 holding other X-vars constant
- β_2 : partial population slope, change in Y per unit change in X_2 holding other X-vars constant
- β_p : partial population slope, change in Y per unit change in X_p holding other X-vars constant

Lecture 10: Regression parameters

Multiple Regression:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i$$

- ϵ_i : unexplained error - difference bw y_i and value predicted by model ($\hat{y}_i$)
- $NPP = \beta_0 + \beta_1(\text{lat}) + \beta_2(\text{long}) + \beta_3(\text{soil fertility}) + \epsilon_i$

Lecture 10: Regression parameters

Multiple Regression:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i$$

- Estimate multiple regression parameters (intercept, partial slopes) using OLS to fit the regression line
- OLS minimize $\sum (y_i - \hat{y}_i)^2$, the SS (vertical distance) between observed y_i and predicted $\hat{y}_i$ for each x_{ij}
- ϵ estimated as residuals: $\epsilon_i = y_i - \hat{y}_i$
- Calculation solves set of simultaneous normal equations with matrix algebra

Lecture 10: Regression parameters

Regression equation can be used for prediction by subbing new values for predictor (X) variables

- Confidence intervals calculated for parameters
- Confidence and prediction intervals depend on number of observations and number of predictors
 - More observations decrease interval width
 - More predictors increase interval width
- Prediction should be restricted to within range of X variables

Lecture 10: Analyses of variance

Variance - SS_{total} partitioned into $SS_{\text{regression}}$ and SS_{residual}

- SSregression is variance in Y explained by model
- SSresidual is variance not explained by model

Source of variation	SS	df	MS	Interpretation
Regression	$\sum_{i=1}^n (y_i - \bar{y})^2$	p	$\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{p}$	Difference between predicted observation and mean
Residual	$\sum_{i=1}^n (y_i - \hat{y}_i)^2$	$n - p - 1$	$\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - p - 1}$	Difference between each observation and predicted
Total	$\sum_{i=1}^n (y_i - \bar{y})^2$	$n - 1$		Difference between each observation and mean

Lecture 10: Analyses

SS converted to non-additive MS (SS/df)

- MSresidual: estimate population variance
- MSregression: estimate population variance + variation due to strength of X-Y relationships
- MS do not depend on sample size

Source of variation	SS	df	MS
Regression	$\sum_{i=1}^n (y_i - \bar{y})^2$	p	$\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{p}$
Residual	$\sum_{i=1}^n (y_i - \hat{y}_i)^2$	$n - p - 1$	$\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - p - 1}$
Total	$\sum_{i=1}^n (y_i - \bar{y})^2$	$n - 1$	

Lecture 10: Hypotheses

Two Hos usually tested in MLR:

- “Basic” Ho: all partial regression slopes equal 0; $\beta_1 = \beta_2 = \dots = \beta_p = 0$
- If “basic” Ho true, MSregression and MSresidual estimate variance and their ratio (F-ratio) = 1
- If “basic” Ho false (at least one $\beta \neq 0$) MSregression estimates variance + partial regression slope and their ratio (F-ratio)
- will be > 1 - F-ratio compared to F-distribution for p-value

Lecture 10: Hypotheses

Also: is any specific $\beta = 0$ (explanatory role)?

- E.g., does LAT have effect on NPP?
- These Hs tested through model comparison
- Model w 3 predictors X1, X2, X3 (model 1):
- $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \epsilon_i$
- To test Ho that $\beta_1 = 0$ compare fit of model 1 to model 2:
- $y_i = \beta_0 + \beta_2 x_{i2} + \beta_3 x_{i3} + \epsilon_i$

Lecture 10: Hypotheses

- If SSregression of mod1=mod2, cannot reject Ho $\beta_1 = 0$
- If SSregression of mod1 > mod2, evidence to reject Ho $\beta_1 = 0$

- SS for β_1 is $SS_{extra\beta_1} = \text{Full } SS_{\text{regression}} - \text{Reduced } SS_{\text{regression}}$
- Use partial F-test to test $H_0: \beta_1 = 0$:

$$F_{w, n-p} = \frac{MS_{Extra}}{FULL MS_{Residual}}$$

Can also use t-test (R provides this value)

Lecture 10: Explained variance

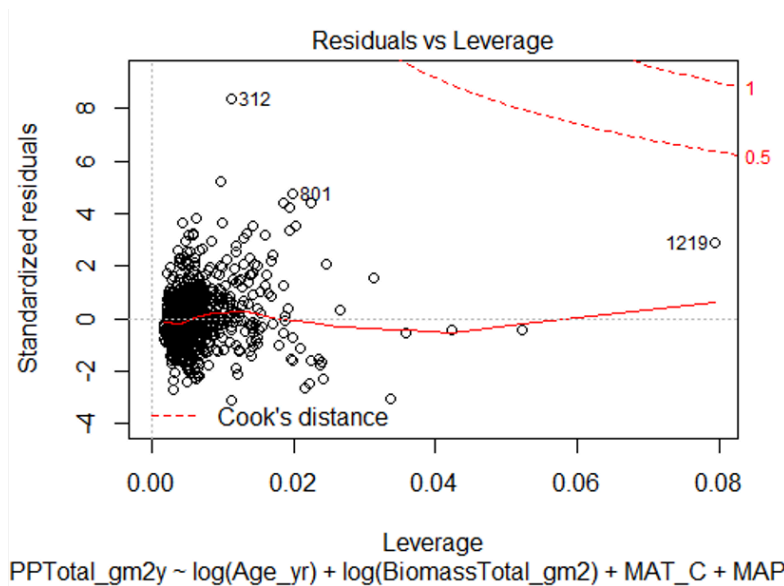
Explained variance (r^2) is calculated the same way as for simple regression:

$$r^2 = \frac{SS_{Regression}}{SS_{Total}} = 1 - \frac{SS_{Residual}}{SS_{Total}}$$

- r^2 values can not be used to directly compare models
- r^2 values will always increase as predictors added
- r^2 values with different transformation will differ

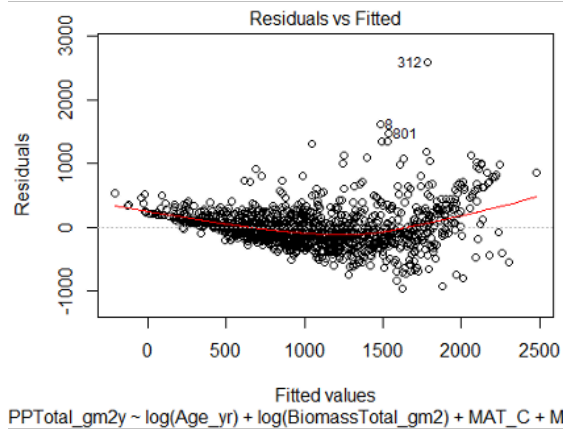
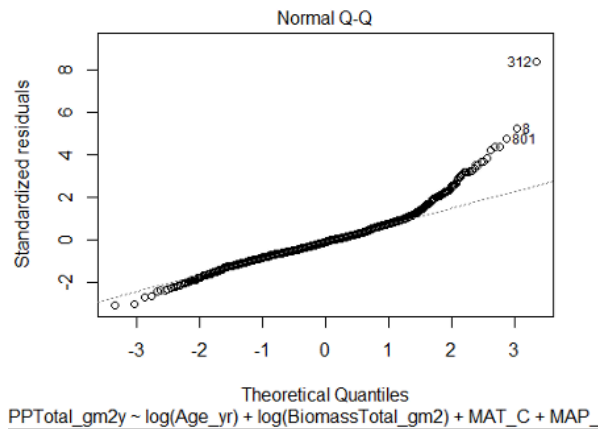
Lecture 10: Assumptions and diagnostics

- Assume fixed Xs; unrealistic in most biological settings
- No major (influential) outliers
- Check leverage, influence- Cook's Di



Lecture 10: Assumptions and diagnostics

- Normality, equal variance, independence
- Residual QQ-plots, residuals vs. predicted values plot
- Distribution/variance often corrected by transforming Y



Lecture 10: Assumptions and diagnostics

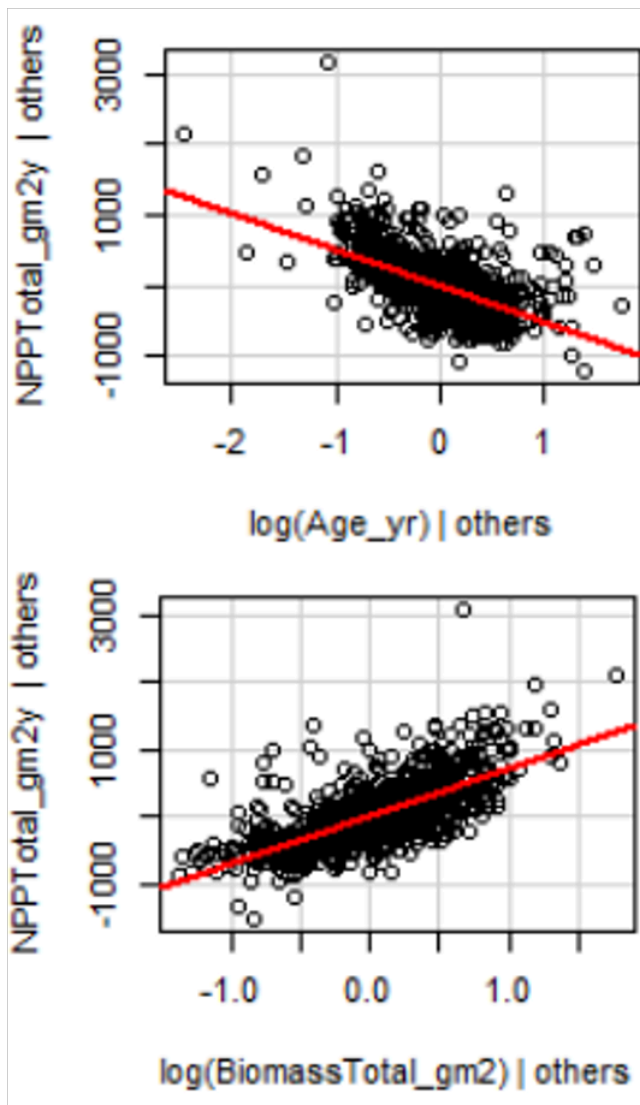
More observations than predictor variables

- Ideally at least 10x observations than predictors to avoid “overfitting”
- Uncorrelated predictor variables (assessed using scatterplot matrix; VIFs)
- Linear relationship between Y and each X, holding others constant (non-linearity assessed by AV plots)

Lecture 10: Analyses

Regression of Y vs. each X does not consider effect of other predictors:

want to know shape of relationship while holding other predictors constant



Lecture 10: Collinearity

- Potential predictor variables are often correlated (e.g., morphometrics, nutrients, climatic parameters)
- Multicollinearity (strong correlation between predictors) causes problems for parameter estimates
- Severe collinearity causes unstable parameter estimates: small change in a single value can result in large changes in β - estimates
- Inflates partial slope error estimates, loss of power

```
> cor(samp[,1:6])
```

	Lat	Long	NPPTotal_gm2y	Lgs_mo	MAT_C	MAP_mm
Lat	1.0000000	-0.02451430	-0.4002104	-0.60704499	-0.70066118	-0.4463274
Long	-0.0245143	1.00000000	-0.1717559	-0.01783002	-0.08775318	-0.2215868
NPPTotal_gm2y	-0.4002104	-0.17175589	1.0000000	0.55946243	0.55177184	0.5278616
Lgs_mo	-0.6070450	-0.01783002	0.5594624	1.00000000	0.92369151	0.5692762
MAT_C	-0.7006612	-0.08775318	0.5517718	0.92369151	1.00000000	0.5788252
MAP_mm	-0.4463274	-0.22158681	0.5278616	0.56927616	0.57882518	1.0000000

Lecture 10: Collinearity

Collinearity can be detected by:

- Variance inflation Factors:
 - $VIF \text{ for } X_j = 1 / (1 - r^2)$

- $VIF > 10 = \text{bad}$
- Best/simplest solution:
 - exclude variables that are highly correlated with other variables
 - they are probably measuring similar
 - thing and are redundant

Lecture 10: Interactions

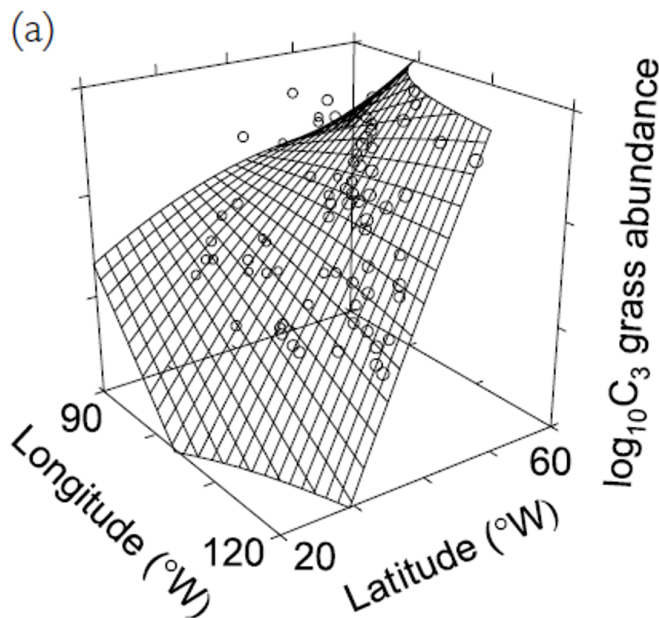
Predictors can be modeled as:

- additive (effect of temp, plus precip, plus fertility) or
- multiplicative (interactive)
- Interaction: effect of X_i depends on levels of X_j
- The partial slope of Y vs. X_1 is different for different levels of X_2 (and vice versa); measured by β_3

$$y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \epsilon_i \quad \text{vs.} \quad y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i1} X_{i2} + \epsilon_i$$

“Curvature” of the regression (hyper)plane

Lecture 10: Analyses



Lecture 10: Analyses

Adding interactions:

- many more predictors (“parameter proliferation”):
- $2n$; 6 params= 64 terms; 7 params= 128
- interpretation more complex
- When to include interactions? When they make biological sense

Lecture 10: Dummy variables

Multiple Linear Regression accommodates continuous and categorical variables (gender, vegetation type, etc.)
Categorical vars as “dummy vars”, n of dummy variables = $n-1$ categories

Sex M/F:

- Need 1 dummy var with two values (0, 1)

Fertility L/M/H:

- Need 2 dummy var, each with two values (0, 1): fert1 (0 if L or H, 1 if M), fert2 (1 if H, 0 if L or M)

Fertility	fert1	fert2
Low	0	0
Med	1	0
High	0	1

Lecture 10: Analyses

Coefficients interpreted relative to reference condition

- R codes dummy variables automatically
- picks “reference” level alphabetically
- Dummy variables with more than 2 levels add extra predictor variables to model

Fertility	fert1	fert2
Low	0	0
Med	1	0
High	0	1

Lecture 10: Analyses

```
> mod1 <- lm(NPPTotal_gm2y ~ MAT_C + MAP_mm + as.factor(Fertility))
> summary(mod1)
```

Call:
lm(formula = NPPTotal_gm2y ~ MAT_C + MAP_mm + as.factor(Fertility))

Residuals:

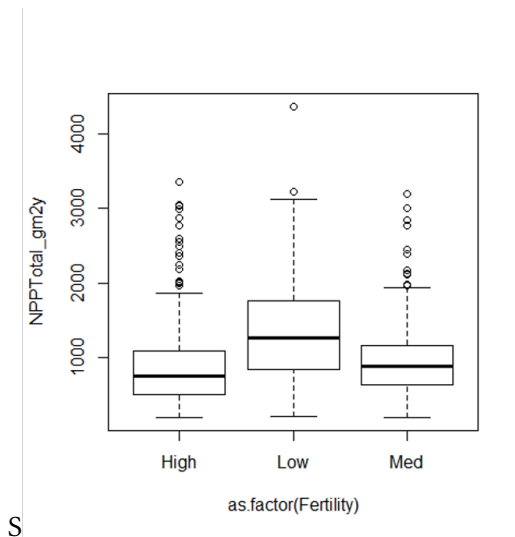
Min	1Q	Median	3Q	Max
-1470.19	-301.59	-49.63	240.82	2901.70

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	359.81714	31.92698	11.270	< 2e-16	***
MAT_C	33.72747	2.73715	12.322	< 2e-16	***
MAP_mm	0.34128	0.03443	9.913	< 2e-16	***
as.factor(Fertility)Low	205.15606	34.36461	5.970	3.1e-09	***
as.factor(Fertility)Med	116.33231	32.30397	3.601	0.000329	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 456.4 on 1232 degrees of freedom
Multiple R-squared: 0.3899, Adjusted R-squared: 0.3879
F-statistic: 196.8 on 4 and 1232 DF, p-value: < 2.2e-16



Lecture 10: Comparing models

When have multiple predictors (and interactions!)

- how to choose “best” model?
- Which predictors to include?
- Occam’s razor: “best” model balances complexity with fit to data

To chose:

- compare “nested” models

Overfitting

- getting high r^2 just by having more (useless predictors)
- so r^2 is not a good way of choosing between nested models

Lecture 10: Comparing models

Need to account for increase in fit with added predictors:

- Adjusted r^2
- Akaike’s information criterion (AIC)
- Both “penalize” models for extra predictors
- Higher adjusted r^2 and lower AIC are better when comparing models

$$\text{Adjusted } r^2 = 1 - \frac{SS_{\text{Residual}}/(n - (p + 1))}{SS_{\text{Total}}/(n - 1)}$$

$$\text{Akaike Information Criterion (AIC)} = n[\ln(SS_{\text{Residual}})] + 2(p + 1) - n \ln(n)$$

Lecture 10: Comparing models

But how to compare models?

- Can fit all possible models
 - compare AICs or adj- r^2 ,
 - tedious w lots of predictors
- Automated forward (and backward) stepwise procedures: start w no terms (all terms), add (remove) terms w largest (smallest)
 - partial F statistic

We will use manual form of backward selection

Lecture 10: Analyses

```
lm(formula = log(NPPTotal_gm2y) ~ log(Age_yr) + log(BiomassTotal_gm2) +  
  MAT_C + as.factor(LeafType))  
  
Residuals:  
    Min      1Q  Median      3Q     Max   
-0.8319 -0.1391 -0.0125  0.1329  1.0132  
  
Coefficients:  
                Estimate Std. Error t value Pr(>|t|)      
(Intercept)      2.106796   0.100738  20.914 < 2e-16 ***  
log(Age_yr)      -0.530852   0.015108 -35.138 < 2e-16 ***  
log(BiomassTotal_gm2) 0.728076   0.013779  52.838 < 2e-16 ***  
MAT_C            0.007861   0.001320   5.954 3.41e-09 ***  
as.factor(LeafType)needle -0.270499   0.014510 -18.642 < 2e-16 ***  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
Residual standard error: 0.2352 on 1215 degrees of freedom    AIC= -62.16  
Multiple R-squared:  0.8059,    Adjusted R-squared:  0.8053
```

Lecture 10: Predictors

Usually want to know relative importance of predictors to explaining Y

- Three general approaches:
- Using F-tests (or t-tests) on partial regression slopes
- Using coefficient of partial determination
- Using standardized partial regression slopes

Lecture 10: Predictors

Using F-tests (or t-tests) on partial regression slopes:

- Conduct F tests of H_0 that each partial regression slope = 0
- If cannot reject H_0 , discard predictor
- Can get additional clues from relative size of F-values
- Does not tell us absolute importance of predictor (usually can not directly compare slope parameters)

Lecture 10: Predictors

Using coefficient of partial determination:

- the reduction in variation of Y due to addition of predictor (X_j)

$$r_{X_j}^2 = \frac{SS_{\text{Extra}}}{\text{Reduced } SS_{\text{Residual}}}$$

SS_{Extra}

- Increased in $SS_{\text{regression}}$ when X_j is added to model
- Reduced SS_{residual} is the unexplained SS from model without X_j

Lecture 10: Predictors

Using standardized partial regression slopes:

- predictors of predictor variables can not be directly compared
- Why?
- Standardize all vars (mean = 0, sd= 1)
- Scales are identical and larger PRS mean more important variable

Lecture 10: Predictors

Using partial r2 values:

Coefficients

	SSR	df	pEta-sqr	dR-sqr
(Intercept)	24.1975	1	0.2647	NA
log(Age_yr)	68.3055	1	0.5040	0.1972
log(BiomassTotal_gm2)	154.4561	1	0.6968	0.4460
MAT_C	1.9615	1	0.0284	0.0057
as.factor(LeafType)	19.2262	1	0.2224	0.0555

Sum of squared errors (SSE): 67.2

Sum of squared total (SST): 346.3

Lecture 10: Reporting results

Results are easiest to report in tabular format

```
lm(formula = log(NPPTotal_gm2y) ~ log(Age_yr) + log(BiomassTotal_gm2) +
  MAT_C + as.factor(LeafType))
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-0.8319 -0.1391 -0.0125  0.1329  1.0132
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    2.106796   0.100738  20.914 < 2e-16 ***
log(Age_yr)    -0.530852   0.015108 -35.138 < 2e-16 ***
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MAT_C          0.007861   0.001320   5.954 3.41e-09 ***
as.factor(LeafType)needle -0.270499   0.014510 -18.642 < 2e-16 ***
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2352 on 1215 degrees of freedom

Multiple R-squared: 0.8059, Adjusted R-squared: 0.8053

Coefficients

```
              SSR df pEta-sqr dR-sqr
(Intercept)  2885.9581  1  0.7174    NA
log(Age_yr)   435.1076  1  0.2768 0.1228
BiomassTotal_gm2 926.7684  1  0.4491 0.2617
MAT_C         64.9419  1  0.0540 0.0183
TEB_DD        16.8202  1  0.0146 0.0047
as.factor(LeafType) 268.0125  1  0.1908 0.0757
```

Sum of squared errors (SSE): 1136.7

Sum of squared total (SST): 3541.9

Lecture 10: Reporting results

Results are easiest to report in tabular format

Response	df	Predictor	coefficient	t	r2	Partial r2	p-value
LN(NPP)	4, 1215	Intercept	2.11	20.9	0.81		<0.0001
		LN(stand age); years	-0.53	-35.1	0.50		<0.0001
		LN(stand biomass); g/m ²	0.73	52.8	0.70		<0.0001
		MAT; °C	0.008	5.95	0.03		<0.0001
		Forest type; needle vs. broadleaf	-0.27	-18.6	0.22		<0.0001